

CodePrep reveals rationale only after an answer is committed

The public demo preserves the interaction; the live path adds authentication, schema validation, and one atomic challenge-and-quota write.

COMMITTED DECISION

CodePrep

technical interview practice

practice

history

repository

user

SESSION 04 / Systems interview

DEMO SESSION

Runs left 47 of 50

Pick one answer. The rationale stays hidden until you commit.

DIFFICULTY

Easy

Medium

Hard

New challenge →

QUESTION 102

MEDIUM

What is the time complexity of finding a value in a balanced binary search tree?

A

$O(1)$

B

$O(\log n)$

Correct

C

$O(n)$

D

$O(n \log n)$

EXPLANATION

Decision accepted.

WHAT THIS TESTS

Balanced-tree search height

Each comparison discards roughly half of the remaining subtree, so the search follows one root-to-leaf path of height $O(\log n)$.

CHALLENGE QUALITY

Useful

Needs work

Default medium fixture · all four options shown · answer B committed · rationale revealed

TWO EXECUTION RAILS

DEMO

Deterministic fixture

Same frontend request contract · in-memory quota · no account, provider, or database

Clerk session + quota remaining > 0

Hosted provider returns one JSON challenge

Pydantic: exactly 4 unique options + answer 0-3

ONE TRANSACTION

Persist challenge + decrement quota

Failure → rollback; quota unchanged

SUPPORTED

Commit-before-rationale interaction and maintained transaction semantics.

NOT CLAIMED

No learning-outcome, interview-performance, question-quality, provider-accuracy, or usage-analytics evidence.